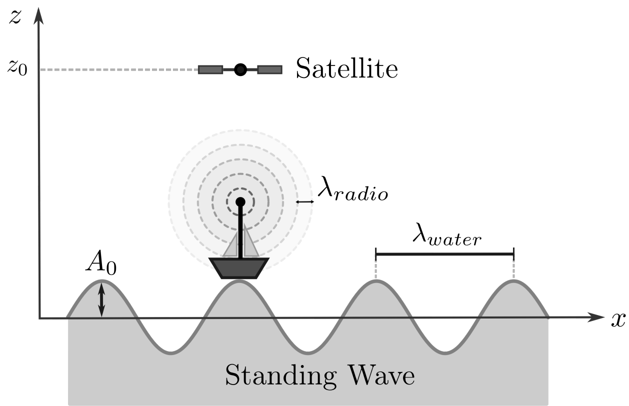
Physik II — FS 2024 — Prof. R. Wallny — Sheet 3

Release: 5. März 2024 – **Due date and discussion:** 14. März 2024

Exercise 1: Boat and satellite

Objective: Wave functions, Doppler effect, energy conservation.

A boat is located on a crest of a harmonic standing wave, as shown in the figure below. The boat is at a fixed position in x ($x(t) = const$) and oscillates in z-direction. The water wave has wavelength λ_{water} and maximum amplitude A_0. The wave velocity (phase velocity) in water is v_{water}. The boat is equipped with a radio station, which emits harmonic electromagnetic waves with wavelength λ_{radio}. Electromagnetic waves propagate with the speed of light $c_{\text{radio}} = 3 \times 10^8\,\text{m/s}$.



a) Write down the displacement of the water wave as a function of time and position. Give in particular the displacement at the position of the boat as a function of time. The boat has a maximal deflection at $t_0 = 0$

b) A geostationary satellite, located directly above the boat, receives the emitted radio waves. Why is the frequency of the received wave not constant? At which position and velocity of the boat (in vertical direction) is the received frequency maximal?

Hint: You will learn in a few weeks that radio waves don't need a medium to propagate. Ignore that fact for this exercise and consider the atmosphere as a medium for those waves.

c) Calculate the maximum frequency that the satellite receives. Write the result as a function of λ_{radio}, c_{radio}, λ_{water}, v_{water} and A_0. Give the numerical result for $\lambda_{\text{radio}} = 3\,\text{m}$, $\lambda_{\text{water}} = 100\,\text{m}$, $v_{\text{water}} = 12\,\text{m}\,\text{s}^{-1}$ and $A_0 = 2\,\text{m}$. Compare the received frequency to the original frequency ν_{radio} emitted by the ship.

d) Calculate the average energy flux density (Poynting vector) of the emitted radio wave at the satellite, assuming that the radio station emits spherical waves isotropically in 3D with an average power of P.

Exercise 2: Coulomb's law

Objective: Application of Coulomb's law.

Two small metal spheres with negligible thickness carrying a charge Q_1 and Q_2 respectively are placed at fixed positions $P_1 = (-1, 0)$ and $P_2 = (1, 0)$ of a two-dimensional coordinate system. The unit of distance is the µm.

a) Give an expression for the force on a test charge $Q_3 = e = |q_{\text{elektron}}|$, located at $P_3 = (3, 3)$, as a function of Q_1 and Q_2.

b) The force of attraction between the two spheres is measured to be equal to 1.08 nN. The spheres are then connected through a thin conductive wire. The connection is then removed and a repulsive force of 0.36 nN is measured between the spheres. What were the initial charges of the spheres?

c) How many electrons flowed from one sphere to the other during the electrical connection?

Exercise 3: Spherical shells

Objective: Gauss' law

Consider two non-conductive spherical shells of respective radii r_A, r_B ($r_A < r_B$) and of respective total charge q_A and q_B. The charges are distributed uniformly over the surface of each shell. The shells are arranged in a concentric manner (the smaller one within the larger one).

a) What is the electric field within the space delimited by the small shell ($r < r_A$)? Justify your answer thoroughly.

b) Compute the electric field outside of the large shell ($r > r_B$).

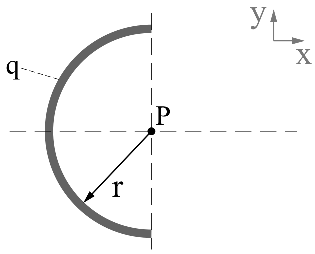
c) How should the ratio of the charges q_A/q_B and their relative signs be chosen in order to have a vanishing electric field for $r > r_B$.

d) Draw the electrical field lines in the case of a vanishing electric field in the region $r > r_B$ and for a positive value of q_A (a sketch of the cross-section through the spherical shell in 2D is sufficient).

Exercise 4: Semicircular Rod

Objective: Continuous charge distribution, Coulomb's Law.

A semicircular rod has a total charge $q = 10^{-9}\,\mathrm{C}$. The charge is distributed homogeneously on the rod. Assume that the rod has a negligible thickness and that its curvature has the radius $r = 1\,\mathrm{m}$.



Determine the electrical field $\vec{E}$ at the centre P of the rod.
Hint: *Exploit the symmetry of the charge distribution and use polar coordinates.*

Exercise 5: Oscillation of a charged particle

Objective: Electric fields, repetition of oscillations.

A ring at rest with radius r_R lies in the y-z plane and carries a positive charge q that is evenly distributed over its length.

a) Show that for $x \ll r_R$ the electric field along the ring axis is proportional to x.

Hint: *In general for $a \ll b$, the approximation $\left(1 + \frac{a}{b}\right) \approx 1$ can be applied.*

b) A point-like particle with mass m and negative charge $-q$ is located along the ring axis. Determine the force on the particle as a function of x, where $x \ll r_R$ still applies.

c) The particle is initially located at the center of the ring. Show that, after a small displacement in positive x-direction, the particle undergoes a harmonic oscillation.

d) What is the frequency of that oscillation?